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REGULAR PAPER

Grasp Planning for Underactuated Three-Finger Grippers Using a Novel Triangular Representation

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Abstract
Robot grasping is a critical area of research in robotics, which aims to develop algorithms and strategies that enable robots towards reliable grasping and manipulation of objects in various environments. The ability to grasp and handle objects dexterously has wide-ranging applications in manufacturing, logistics, and healthcare. Recent advances in deep learning have paved the way for learning-based approaches toward autonomous grasping and intelligent manipulation in structured and unstructured environments. However, majority of the existing methods are designed for two-finger robotic grippers and cannot be extended towards application on three-finger robotic grippers. Three-finger grippers can perform a wide variety of grasps, and thereby require more nuanced grasp representation as compared to existing approaches. In this work, we propose a novel method towards representing underactuated three-finger grasps. A new grasp data set is also presented, specifically designed to train and evaluate three-finger robotic grasps using the proposed novel representation. A state-of-the-art deep learning architecture was trained on the dataset towards developing an end-to-end grasp planning pipeline. The proposed pipeline uses color and depth images as inputs to predict the ideal grasp pose for a three-finger robotic gripper to grasp the target object. The trained network was validated through simulation and hardware experiments, and the results were compared with an existing state-of-the-art grasp planning approach to highlight the potential of this new technique for grasp representation and planning.

Keywords Grasp planning · Three-finger gripper · Deep learning · Grasp dataset generation

1 Introduction

Intelligent grasping has always been a fundamental requirement in translating robotics research towards real-world applications, such as material handling and assembly. It is especially important for handling harmful materials and/or working in a hazardous environment. One example is the semi-autonomous robotic system for recycling Li-ion batteries proposed by S. Gadgil et al. [1]. Several algorithms have been developed for synthesizing robotic grasps in order to achieve stability, force-closure, task compatibility and other desirable characteristics. These algorithms can be broadly classified into analytical and empirical/data-driven approaches. Analytical approaches are those based on geometric, kinematic and/or dynamic formulations of the grasp synthesis problem. The empirical/data-driven approaches avoid the need for explicit mathematical modeling by learning or imitating grasps provided through datasets.

Bicchi and Kumar [2] have given a comprehensive survey of the field of robotic grasping, focusing on the development of theoretical frameworks and analytical approaches where they also distinguish between the tasks of restraining objects, referred to as fixturing, and manipulating objects with fingers, referred to as dexterous manipulation. For most industrial pick-and-place tasks, restraining an object or fixturing is enough. Only a few complex tasks will require dexterous in-hand manipulation. The pick and place tasks can be further classified based on various types of grasps, including enveloping grasps, which are superior for object restraint, and fingertip grasps, used for fine manipulation.

Analytical approaches tend to be robust, but computationally expensive. These methods typically require precise knowledge of object properties and contact mechanics, which may not be readily available in practical applications. Recent

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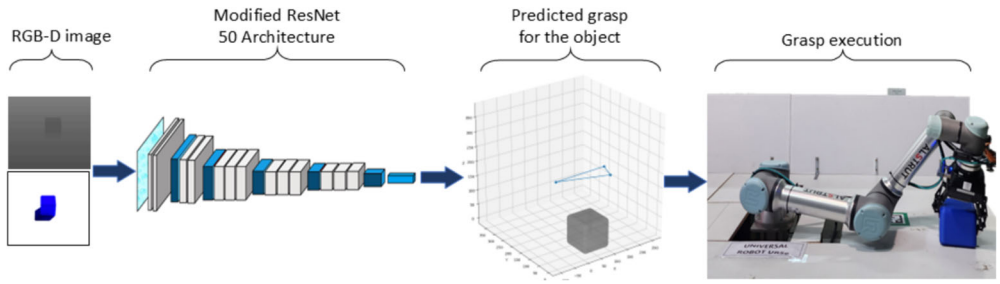

Fig. 1 Proposed grasp planning pipeline for underactuated three-finger grippers using novel triangle representation

advancements in deep learning have paved the way for learning-based approaches to be applied in autonomous grasping and intelligent manipulation. In this context, several datasets and methodologies have been proposed, primarily for two-fingered grippers. Among these, the Cornell dataset [3] and the Jacquard dataset [4] have gained prominence due to their ease of use, as they rely solely on RGB-D (red-green-blue-depth) image input and provide fast grasp predictions for two-finger grippers. However, these datasets employ rectangle-based criteria for generating suitable grasps, which is not optimal for three-finger grippers, as they can offer diverse grasp configurations when compared to a two-finger gripper.

To address the limitations encountered while using deep learning methods for grasp planning for underactuated three-fingered grippers, we propose:

1. A novel grasp representation tailored for underactuated three-fingered grippers: SE(3)[1]+ triangle.
2. New dataset specifically for underactuated three-fingered grippers using the proposed novel grasp representation.
3. A modified ResNet50 model capable of predicting ideal grasp pose in the proposed representation, based on color and depth data.
4. Real-world evaluation of the grasp performance for an end-to-end deep neural network trained over the new dataset, as shown in Fig. 1.

2 Review of Existing Grasp Planning Methodologies

As mentioned in the previous section, grasping can be posed as an analytical problem. A good grasp requires force balancing for stability. Early works on grasp synthesis focused on providing analytical solutions based on mechanics, which are subdivided into two types: force closure and form closure [5].

[1] SE(3) refers to the Special Euclidean (3) group which is a collection of all physically possible translations and rotations for a rigid body in 3D space.

Force closure refers to the situation where a grasp is stable based on the forces applied to the object by the gripper. Form closure, on the other hand, refers to the situation where grasp stability is ensured by constraining the object using geometrical constraints. In other words, the object is grasped in such a way that no matter how the object is loaded or manipulated, it will not slip out of the gripper. Bichhi et al. [2, 5] show that form closure is solved as a geometric problem and force closure as a force constraint problem between the object and the gripper. A robotic grasping simulator called GraspIt! [6] is available to make grasp analysis and planning easier. It allows for interactive evaluation and visualization of grasps by importing different hand and object models. Some of the important aspects of GraspIt! are its collection of robot models, contact determination, collision detection, and dynamic simulation. In general, GraspIt! is a valuable tool for robotic manipulation research and development, as it allows for the verification of gripper designs and evaluation of their performance in a simulated environment before deploying them on actual robots.

Analytical grasp synthesis is usually formulated as a constrained optimization problem that tries to solve for one or several of the following:

1. Grasping points on the object
2. The approach vector describing the 3-D angle with which the gripper approaches the grasp point
3. The wrist orientation of the gripper
4. An initial finger configuration of the gripper

However, these analytical methods often require simplification of physical models, leading to unrealistic results. They also require a fully observable environment, including knowledge of object and gripper properties, such as object weight, stiffness, and contact friction model, which may not be available in real-world scenarios.

With the rapid development and popularization of learning techniques, data-driven approaches are gaining dominance in multiple domains such as autonomous driving [7, 8], robot navigation [9, 10], and grasp planning applications [11, 12]. A detailed survey of data-driven grasp synthesis was done by Jeannette Bohg et al. in [13]. Data-driven approaches can be

further classified into imitation-based methods and end-to-end learning-based methods. In imitation-based methods, a dataset of stable force-closure grasp is labelled for predefined simple geometries. Any novel object geometry is then compared to this dataset and a predefined grasp is assigned for the object. This method is demonstrated by Zeng deg et al. [14], where one can only select grasps from predefined options. End-to-end learning implies that all sensor data (including images, depth data, and tactile sensor data) are provided to a deep neural network, which then predicts an ideal grasp configuration.

End-to-end learning is gaining popularity as it provides reliable grasps compared to imitation-based methods and is easier to deploy. Ian Lenz et al.[15] demonstrated a method to predict a rectangular representation of grasps for a two-finger gripper using a two-stage cascaded deep learning detection model that generated grasps in the rectangular representation and ranked them. S. Kumra et al. [16] used this rectangular representation with a Convolutional Neural Network (CNN) based approach to achieve an accuracy of 89.21% on the Cornell Grasp Dataset. Zhang et al. [17] used a CNN-based approach to estimate the most effective grasping point for a vacuum gripper.

Some work [18, 19] has also been done in utilizing data in the form of point clouds using PointNet [20] and PointNet++ [21]. A common problem with point cloud data is that objects detected are incomplete since the sensor is usually a camera or LiDAR. W. Chen et al. [22] proposed an architecture with a transformer-based encoder and manifold-based decoder for shape completion that combats this drawback. DC. Hoang et al. [23] used an attention mechanism along with PointNet++ to effectively capture relationships in data across channel, space, and time.

The end-to-end learning based approach simplifies the problem and can thereby lead to faster solutions when compared to analytical methods, but it is limited only to two-finger grippers.

Some of the major robotic grasp datasets are summarised below:

1. Cornell Grasping Dataset [3]: Contains 885 RGB-D images of household objects, each annotated with multiple successful and unsuccessful grasps, including depth information for 3D grasp planning.
2. DexNet 2.0 [11]: A synthetic dataset with over 6.7 million grasps, using physics-based analysis and the Ferrari-Canny metric to label grasps. Includes depth images and analytic grasp quality metrics.
3. DexNet 3.0 [24]: Builds on DexNet 2.0 with over 1 million grasps, focusing on grasp robustness and efficiency through deep imitation learning. Provides multi-view depth images and 3D mesh models.
4. Jacquard Dataset [4]: A large-scale synthetic dataset with 54,000 RGB-D images and over 1 million grasps, designed for deep learning-based grasp detection.
5. VMRD [25]: Focuses on multi-object manipulation tasks with annotations for grasp poses and object-object relationships, supporting relational reasoning in robotic grasping.
6. GraspNet-1Billion [26]: A real-world dataset with 1 billion annotated grasps over 88 objects, emphasizing 6 degrees-of-freedom(DoF) robustness and diversity in grasp planning.
7. ACRONYM [27]: A synthetic dataset with over 17 million grasps for thousands of 3D models, suitable for simulated and real-world grasp planning tasks.
8. REGRAD [28]: Focuses on real-time grasp adaptation with sensor data and dynamic adjustments during grasping tasks, supporting research on adaptive grasping techniques.

The datasets [3, 4, 11, 25, 26, 28] use a rectangular representation. The rectangular representation was first introduced by Yun Jiang et al. [3] and is primarily for two-fingered grippers/parallel jaw grippers. The visualization of the rectangular grasp is shown in Fig. 2. The correlation between the gripper properties (jaw height, jaw width, approach direction) and rectangular grasp representation for an object is explained as follows. In Fig. 2, the spoon is the object to be grasped using the two-finger gripper. The rectangle representation uses five dimensions to represent a grasp

$$G = \{x, y, \theta, w, h\}$$

where $\{x, y, \theta\}$ are the desired position and orientation of the gripper as shown in Fig. 2. The w, h are the desired jaw width and jaw height for the gripper to grasp the object.

While the industry primarily relied on systems using two-finger grippers, recently, there has been more interest towards the use of three-finger and n-finger grippers. In comparison with two-finger grippers, three-finger grippers like the Robotiq three-finger Adaptive Gripper, Barrett hand[29] can do both enveloping grasps as well as fingertip grasps due to their adaptability and versatility.

But the rectangular representation described above works well for a two-finger gripper or parallel jaw gripper, it is not applicable to n-finger grippers, including three-finger grippers, because of the following reasons:

- Rectangular notation is mainly useful for planar (2D) object picking with two-finger grippers. (SE(3) + Rectangle representation as used by GraspNet-1Billion [26] and REGRAD [28] is not suitable for three-finger grippers as we lose triangle vertex data)

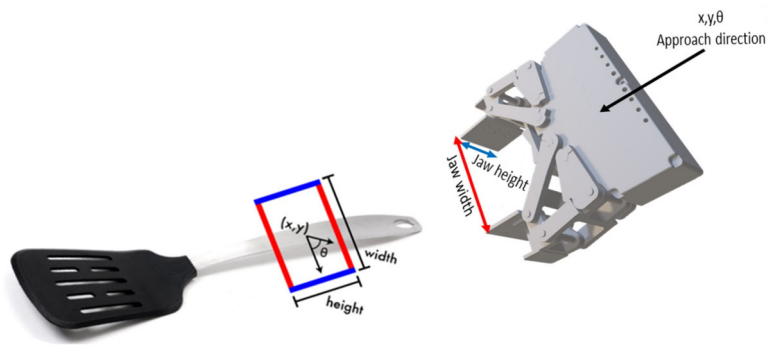

Fig. 2 Rectangular grasp representation for a two finger gripper trying to grasp an object

- Three-finger grippers have different grasp modes such as pinch, spherical or encompassing grasp, suited for different categories of objects.
- Most n-fingered grippers have adduction/abduction capabilities to their finger in addition to flexion/extension. The rectangular notation cannot represent these additional degrees of freedom.

3 Proposed Grasp Representation

To address the limitation of existing rectangular representation to denote ideal grasps for robotic grippers that use three fingers, this work proposes a novel SE(3) + triangle representation.

A good grasp representation should encompass both the spatial pose of the gripper base/palm as well as the pose of the gripper fingers relative to the object being grasped or with respect to global coordinate frames. For an underactuated three-finger gripper, the finger configuration for a grasp could be represented using an isosceles triangle, with each vertex representing the tip of a finger. Combined with the pose of the gripper palm, this can be given as

$$G = \{x, y, z, \theta_x, \theta_y, \theta_z, H, W\}$$

wherein:

- $[x, y, z]$ are the spatial coordinates of the gripper's palm in three-dimensional space.
- $[\theta_x, \theta_y, \theta_z]$ denote the orientation(roll, pitch, yaw) of the gripper palm from it's initial configuration.
- $[H, W]$ captures the base of the triangle and height of the triangle formed by the three finger tips of the gripper.

The graphical representation for the proposed grasp notation is shown in Fig. 3.

3.1 Proposed Grasp Representation as Applied to Robotiq Three-Finger Gripper

An isosceles triangle in 3D space can be denoted by 8-dimensional notation: $[x, y, z, \theta_x, \theta_y, \theta_z, H, W]$. But, for an underactuated gripper such as the three-finger Robotiq gripper and Barrett hand this could be denoted using a 7-dimensional configuration instead of 8. Since the underactuated three-finger grippers are symmetric, the geometric correlation between H and W allows us to decrease the dimensionality to 7, as given by:

$$G = \{x, y, z, \theta_x, \theta_y, \theta_z, W\}$$

or

$$G = \{x, y, z, \theta_x, \theta_y, \theta_z, H\}$$

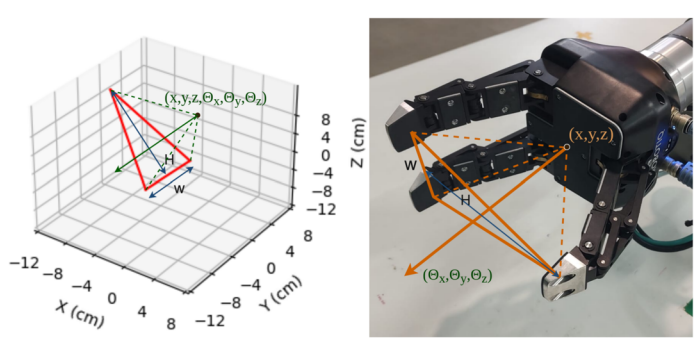

Fig. 3 Proposed triangular notation overlapped on three-finger Robotiq gripper

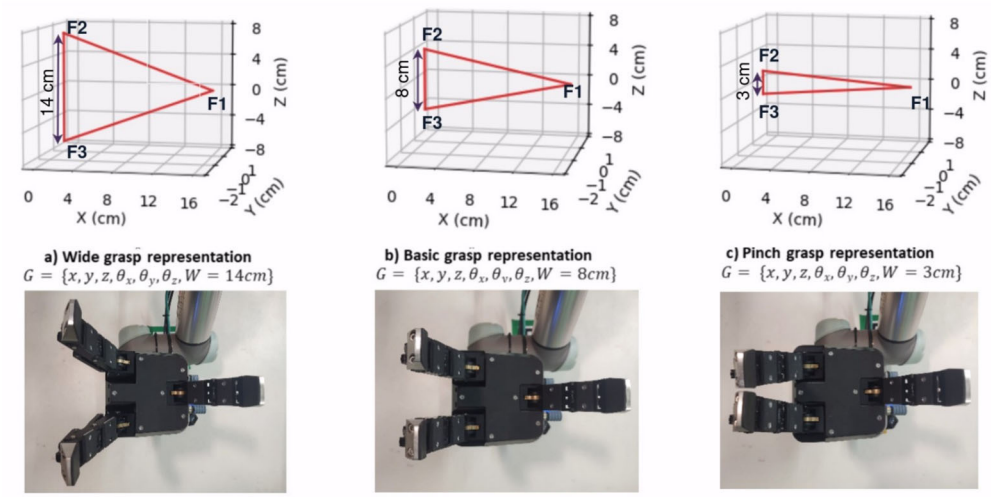

Fig. 4 Triangular grasp representation for Robotiq three-finger gripper under (a) wide, (b) basic, and (c) pinch grasp mode

For the Robotiq three-finger gripper, the distance between finger 1 and the line joining finger 2,3, given by H, is constant. As such, the following representation is used, $[G = x, y, z, \theta_x, \theta_y, \theta_z, W]$. The different grasp modes for the Robotiq-three-finger gripper and the corresponding grasp notations are shown in Fig. 4:

- Wide Grasp Representation (a): This representation features a wide grasp where the fingers are spread apart, with a grasp width(W) of 14 cm.
- Basic Grasp Representation (b): This is a standard grasp with a moderate width(W) of 8 cm. The fingers are closer together compared to the wide grasp, allowing for a more secure grip on moderately sized objects.
- Pinch Grasp Representation (c): The pinch grasp has the fingers very close together with a width(W) of 3 cm, suitable for grasping smaller or thinner objects.

Note that these are the standard grasp modes for the Robotiq three-finger gripper. It is possible to interpolate in-between positions by varying grasp width(W).

While the rest of this manuscript validates the proposed novel grasp approach on the Robotiq three-finger gripper, it is possible to use the proposed grasp representation for other robotic grippers such as the Barrett hand as well.

4 Dataset Creation

This section describes the development of a dataset that includes sensor data corresponding to the object to be grasped as well as the ideal grasp for the object, using the proposed novel grasp representation. To reduce the manual effort involved in creating the dataset, the sensor data for the object, including RGB image and point cloud data, would be generated through rendering for the chosen object set. The ideal grasp for the object is generated using GraspIt!.

4.1 Object List and Sensor Data Generation

Based on a review of existing graspable object datasets, the Extended Grasping Analysis Dataset (EGAD) [30] was chosen for the list of objects to train and validate grasping. EGAD is a state-of-the-art dataset consisting of a diverse collection of objects designed to test the robustness and versatility of grasping algorithms. The dataset includes a total of 2330 objects, created using generative design methods. These objects are categorized on various geometric and physical properties to challenge grasp planners over different scenarios. Of the 2330 shapes, 2281 are meant for training and 49 are for validation.

As shown in Fig. 5, the 2281 training and 49 validation objects vary in their grasp difficulty and shape complex-

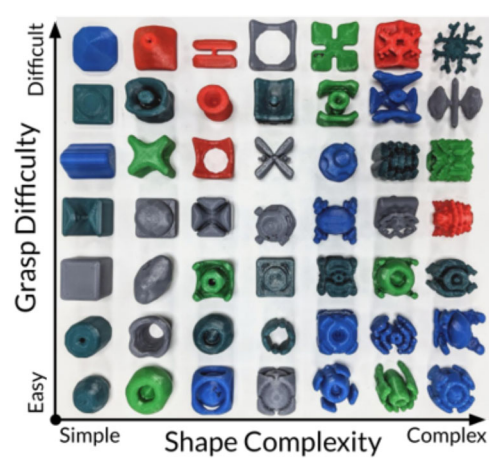

Fig. 5 Shape variation within the validation set of Extended Grasping Analysis Dataset (EGAD)[30]

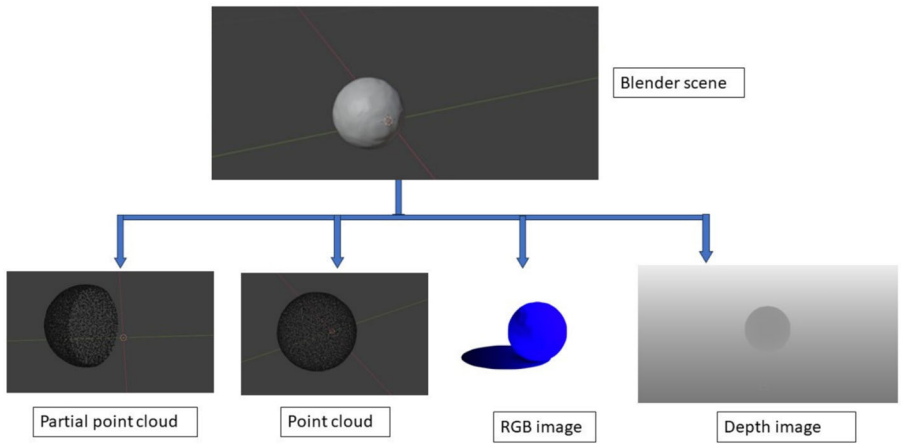

Fig. 6 Blender setup for synthetic sensor data generation: The object model from the EGAD dataset is shown in the first row. The partial and full point cloud, RGB image and depth image for the object obtained from Blender are shown in the second row

ity dimension. The X-axis shows shape complexity and Y-axis shows grasp difficulty, going from easy/simple to difficult/complex. This provides us with a good variation of objects to evaluate the robustness of grasp planning.

Another advantage of the EGAD dataset is that the CAD models of all the objects in the dataset are readily available. This makes it easy for research communities across the world to validate their trained grasp planning architectures over real-world experiments and compare results.

The object size in the EGAD dataset can be varied depending on the gripper size. In our work, we modified the objects size to the appropriate size for 12 cm wide antipodal grasp, resulting in a training dataset consisting of 2281 objects and a validation dataset comprising 49 objects, with required dimensions.

As mentioned earlier, the sensor data for the chosen objects, including photorealistic RGB(red, green, blue) images, along with partial and full point clouds were generated using an open source 3D rendering software, Blender [31], making it scalable if more objects are to be added in future. To achieve photorealistic rendering, the intrinsic parameters of an Intel RealSense D455 camera were used, as given below:

- Image width: 640 pixels
- Image height: 480 pixels
- Principal point: (320, 240) pixels
- Focal length: 387 mm

The above parameters were chosen as the same camera was later used in hardware validation. Figure 6 shows the setup for synthetic data generation using Blender.

4.2 GraspIt!

GraspIt! [6] is a platform to evaluate and plan grasps with various robotic hands and object models. It allows for testing and refining grasps in a simulated environment before deploying them on physical robots.

GraspIt! supports a wide variety of robotic hands, including models like the Barrett hand, DLR hand, and Robonaut hand, along with object models and tools for grasp quality evaluation.

For any given object model with material properties, GraspIt! simulates multiple grasp configurations, evaluating them based on stability and contact points. Based on the results, GraspIt! selects the ideal grasp by maximizing grasp quality based on the Grasp Wrench Space (GWS) and the Epsilon Quality Metric (ϵ). GWS represents the set of all possible wrenches (force and torque combinations) that the grasp can apply to an object:

$$\mathcal{W} = \left\{ \sum_{i=1}^{n} \lambda_i \mathbf{w}_i \;\middle|\; \lambda_i \geq 0 \right\} \tag{1}$$

where w_i are the primitive wrenches at each contact point, and λ_i are non-negative scaling factors.

The epsilon metric ϵ is defined as:

$$\epsilon = \min_{\|\mathbf{f}\|=1} \max_{\mathbf{w}\in\mathcal{W}} \mathbf{f} \cdot \mathbf{w} \tag{2}$$

where $\mathbf{f}$ represents an external unit wrench (force and torque) that the grasp must resist.

4.3 Generating Ground Truth Grasps

The ideal grasp poses for all objects in the new dataset was generated using the GraspIt!. The 10 best grasp poses generated by GraspIt! serve as the ground truth for training the models. The dataset generation process is shown in Fig. 7.

The object to be grasped is added to a scene, which includes a table and three obstacles as shown in Fig. 7. The obstacles help limit the grasp sampling to be within the reachable workspace of the robotic manipulator that moves the

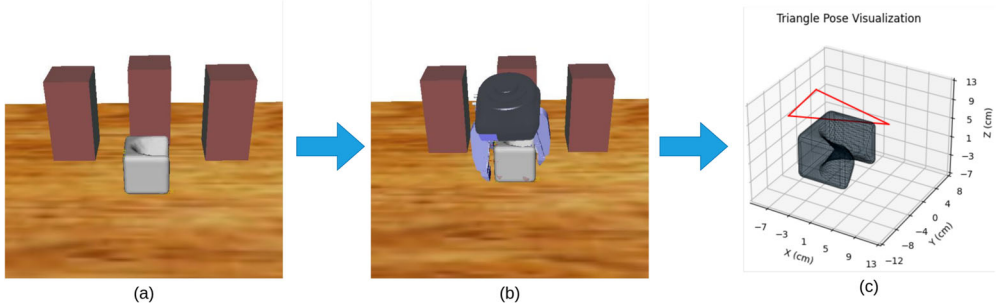

Fig. 7 Dataset creation using GraspIt!: (a) The object to be grasped (b) The ideal grasp generated by GraspIt!, while avoiding collision with table and other objects (c) The ideal grasp in proposed triangular representation, overlaid with the object

gripper. It also ensures that the ideal grasps generated are inclined towards the camera. The best ten grasps generated by GraspIt! and represented in the novel format explained in Section 3 is added to the dataset corresponding to the object data in the following tabular format, showing one of the stored instances: (Tables 1 and 2)

The recorded data in tabular format provides the grasp quality and the gripper pose and configuration for each grasp.

- X, Y, Z: The Cartesian coordinates of the origin of the gripper in the GraspIt! reference frame.
- Roll, Pitch, Yaw: The orientation of the gripper in the GraspIt! reference frame.
- Lateral Joint 1, Lateral Joint 2: These represent the configurations of the lateral joints of the Robotiq three-finger gripper that contribute to the grasp configuration.
- Epsilon Quality: This metric quantifies the proposed grasp's ability to resist external forces in any direction, with higher values indicating more stable grasps.
- Volume Quality: This metric measures the volume of the grasp polyhedron. Larger volumes correspond to better grasps.

4.4 Three-finger Grasp (3FG) Dataset

Table 3 presents a comparison between existing robotic grasp datasets and the new three-finger grasp (3FG) dataset created as part of this work. As mentioned above, the novel 3FG dataset contains simulated RGB and point cloud data as recorded by a virtual camera in Blender. The rendered images and point clouds include only a single object per scene. The novel Triangle + SE(3) representation proposed in Section 3 was used to record ten suitable grasps per object. Each suitable grasp were generated using GraspIt!, allowing for the ideal grasp poses as well as grasp quality to be recorded. The overall dataset includes 2281 objects and the corresponding best ten ideal grasp pairs. The simulation environment is shown in Fig. 7 and the recorded data has the format shown in Section 4.3. At the time of preparation of this manuscript, there are no publicly available datasets specifically designed for three-finger grippers. The proposed dataset aims to assist researchers and engineers in developing and testing solutions, while also contributing to the standardization of data for three-finger robotic grasping research.

Table 1 Grasp position and orientation

X	Y	Z	Roll	Pitch	Yaw
0.11462	0.03071	0.20608	-1.45105	-0.75136	2.76147

5 Learning for Grasp Planning Using the Triangular Representation

This section details the training and validation of a modified ResNet50 architecture on the new 3FG dataset for grasp planning.

5.1 Network Architecture

ResNet50, introduced by He et al. [32], is a deep convolutional neural network, notable for its residual connections that alleviate the vanishing gradient problem. As the name indicates, the architecture contains 50 layers, including convolutional, batch normalization, ReLU, and pooling layers, divided into 5 stages, each with a series of convolutional blocks.

Considering the input sensor data from the 3FG dataset includes RGB-D images, the first convolutional layer of ResNet50 had to be modified to accept four input channels instead of three. The modified convolutional layer has a kernel size of 7 × 7, stride 2, and padding 3. The backbone

Table 2 Joint configuration and quality metrics

Lateral Joint 1	Lateral Joint 2	Epsilon Quality	Volume Quality	W
0.03259	-0.03259	0.210519	1.267796	12

extracts a high-dimensional feature vector, which is then passed through three fully connected branches. Each branch is composed of four fully connected layers with decreasing units to predict the seven-dimensional ideal grasp parameters, given by

$$G = \{x, y, z, \theta_x, \theta_y, \theta_z, W\}$$

The structure and functionality of each of the three branches are explained below:

- Centroid prediction: The first branch predicts the gripper's 3D centroid (x, y, z). The four fully connected layers for centroid prediction have 2048, 1024, 512, and 3 units, respectively, with the final layer providing the 3D coordinates.
- Orientation prediction: The second branch predicts the roll, pitch, and yaw angles $(\theta_x, \theta_y, \theta_z)$. This branch also consists of 2048, 1024, 512, and 3 units, with the final layer predicting three orientation angles.
- Finger distance prediction: The third branch estimates the distance between two fingers of the gripper. This branch uses 2048, 1024, 512, and 1 unit for predicting the scalar distance W.

The modified ResNet50 model capable of predicting the ideal grasp pose in the above format, based on four-channel RGB-D inputs is shown in Fig. 8 and a detailed block diagram for same is shown in Fig. 9

5.2 Loss Function Formulation

The modified ResNet50 model was trained to minimize the composite loss function given below:

$$\text{Loss} = \sum_{i=1}^{3} \mathcal{L}_{\text{pos}_i} + \mathcal{L}_{\text{centroid}} + \mathcal{L}_{\text{orientation}} + 0.001 \cdot \mathcal{L}_{\text{distance}}$$

The loss function combines errors from the predicted vertex positions, centroid, orientation, and finger distance. The distance component is scaled to 0.001 due to its relatively larger magnitude compared to the orientation and grasp width components. The lower weight ensures balanced gradient contributions during training. Each loss term is calculated using the mean absolute error

$$\mathcal{L} = \frac{1}{n} \cdot \sum \left[|x - \hat{x}| + |y - \hat{y}| + |z - \hat{z}| \right]$$

where $\mathcal{L}_{\text{pos}_i}$ represents the loss for each vertex, $\mathcal{L}_{\text{centroid}}$ for the centroid of the gripper, $\mathcal{L}_{\text{orientation}}$ for its orientation, and $\mathcal{L}_{\text{distance}}$ for the distance between the fingers W.

The model's performance was evaluated using Root Mean Square Error (RMSE) and the R^2 score. RMSE provides the average prediction error, calculated as the square root of the mean squared differences between predicted and actual values. RMSE also serves as a comparison metric, as it directly represents the Euclidean distance between the predicted and actual positions. The R^2 score for the predicted vertex

Table 3 Comparison of existing grasp datasets with proposed 3FG dataset

Dataset	Representation	Modality	Source	Size	Object/Scene	Grasp/Scene
Cornell [3]	Rect.	RGB-D	Real	1035	1	∼8
Dex-Net 2.0 [11]	Rect.	Depth	Simulation	6.7M	1	1
Dex-Net 3.0 [24]	Point	Depth	Simulation	2.8M	1	1
Jacquard [4]	Rect.	RGB-D	Simulation	54k	1	∼20
VMRD [25]	Rect.	RGB	Real	4.7k	∼3	∼20
GraspNet-1billion [26]	Rect. + SE(3)	RGB-D	Real	97k	∼10	3-9M
ACRONYM [27]	SE(3)	Depth	Simulation	8.8k	1	2k
REGRAD [28]	Rect. + SE(3)	RGB-D	Simulation	900k	1-20	1.02k
3FG Dataset	**Triangle + SE(3)**	**RGB-D + point cloud**	**Simulation**	**2.3k**	**1**	**1**

[1]Object/Scene indicates the number of objects present in the workspace at a given time. For example, 1 means there is only one object in the grasp workspace in a given instance.
[2]Grasp/scene indicates the number of grasps generated for objects in the scene.
[3]3FG Dataset is the new dataset created as part of this work.

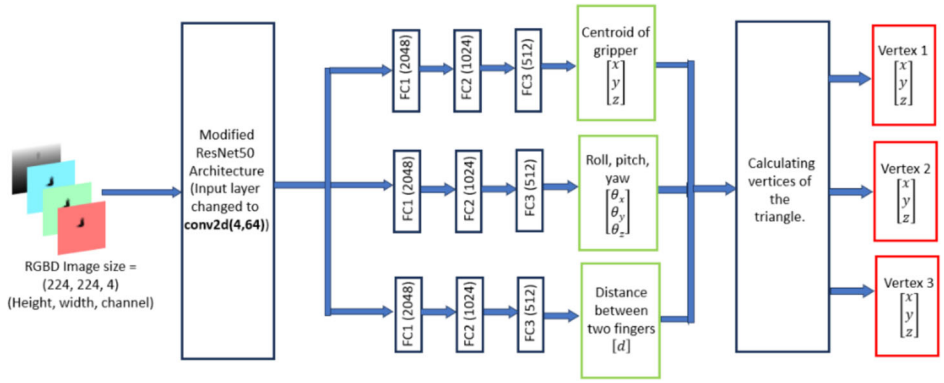

Fig. 8 Overall layout for the modified ResNet50 model

positions measures the proportion of variance in the dependent variable explained by the model, with values closer to 1 indicating better predictive performance and values less than 0 suggesting poor model fit relative to the mean prediction.

5.3 Training Results

The modified ResNet50 model was trained for 50 epochs, after which the loss plateaued, as shown in Fig. 10. The RMSE for the finger positions (triangle vertices) averaged around 2.3 cm as shown in Fig. 11. The R^2 score for the vertex predictions reached 89% as shown in Fig. 12). Figures 10, 11, and 12 show a steady decrease in the loss, along with a decrease in the RMS error as the number of epochs increases. This indicates that the dataset is effective in training the model to a high degree of accuracy.

6 Grasp Planning Validation

In order to compare the grasp performance of the trained architecture with existing state-of-the-art grasp planning approaches, a novel grasp quality metric is defined. The quality of predicted grasps for the object from the EGAD dataset under the trained architecture and existing state-of-the-art grasp planning approaches is compared in a simulated setting. Finally, hardware validation of the trained architecture is also presented in this section.

6.1 Grasp Quality Metric

For a given grasp, the normal force values can be used to compute the wrench at each contact point for each finger:

$$\text{Wrench, } \mathbf{w} = \begin{bmatrix} \mathbf{f} \\ \mathbf{r} \times \mathbf{f} \end{bmatrix}$$

where $\mathbf{f}$ is the force vector and $\mathbf{r}$ is the position vector from the gripper base to the contact point. The grasp wrench space matrix G is constructed by stacking wrenches from all contact points,

$$G = [\mathbf{w}_1 \ \mathbf{w}_2 \ \ldots \ \mathbf{w}_n]$$

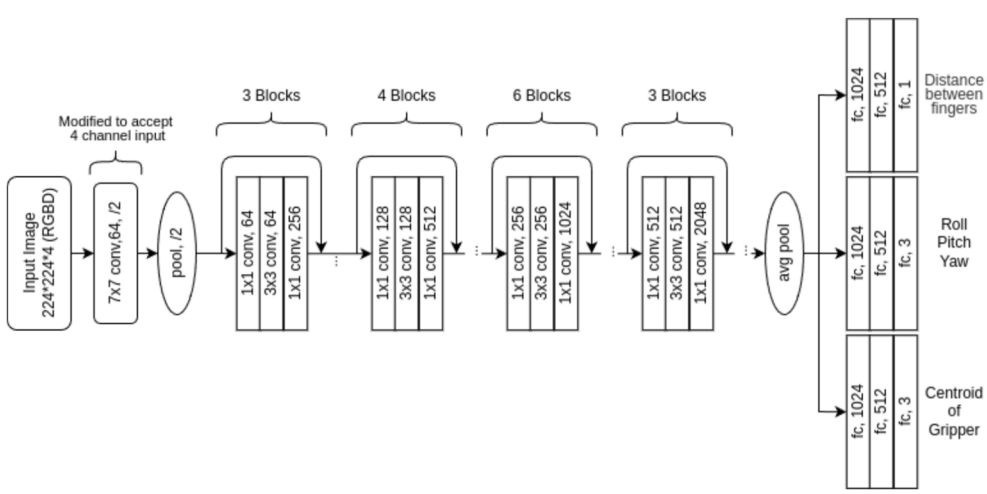

Fig. 9 Modified ResNet50 Network Architecture

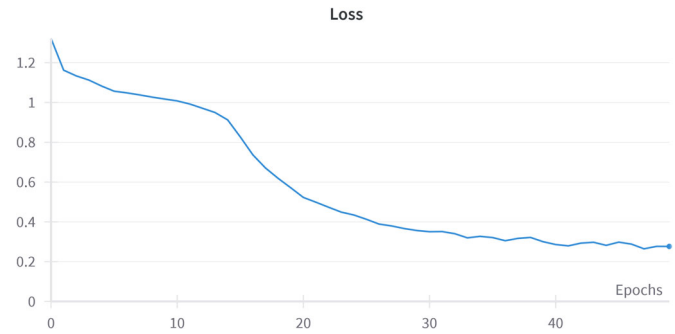

Fig. 10 Loss values over 50 epochs of training

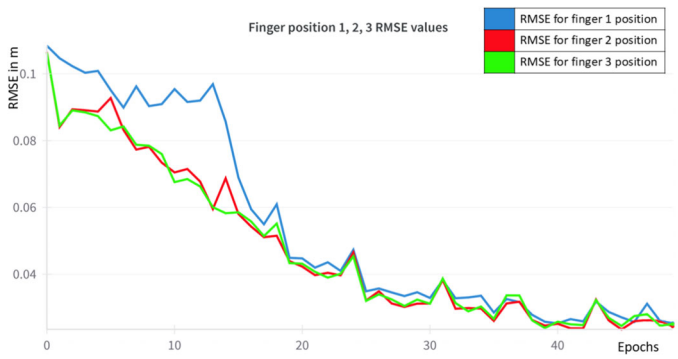

Fig. 11 Finger position Root Mean Squared Error (RMSE) value over 50 epochs of training

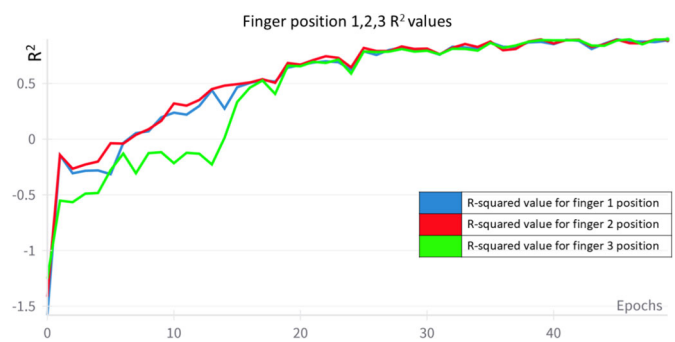

Fig. 12 R-squared values over 50 epochs of training

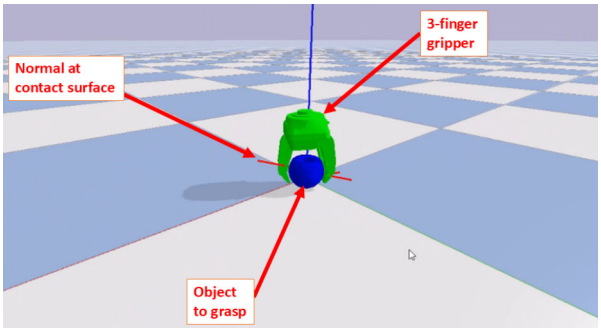

Fig. 13 PyBullet setup for computing grasp quality

Finally, grasp quality metric ϵ is computed as the minimum singular value of G.

$$\epsilon = \min(\sigma(G))$$

A higher minimum singular value, denoted as ϵ, indicates a larger wrench space that the grasp can resist, which directly implies increased stability under external disturbances. This also means that smaller values of ϵ imply vulnerability or grasp instability in the presence of force disturbances. Thus, maximizing ϵ ensures that the grasp can resist perturbations uniformly across directions as explained by Ferrari et al. in [33].

6.2 Simulation Setup for Grasp Quality Comparison

To evaluate the grasp quality of predictions from the trained network as compared to grasp predictions by GraspIt!, a simulation environment was developed using PyBullet. The simulated environment consists of any object from the EGAD dataset and a Robotiq three-finger gripper to grasp the object, as shown in Fig. 13

For a given object, both the trained ResNet50 model and GraspIt! were used to generate ideal grasps. For each ideal grasp pose, the simulated gripper was moved to the desired position and orientation, and grasping action was initiated by closing the fingers. Once a stable grasp is achieved in simulation, the finger contact points and the normal forces are computed from the simulation. The normal force values can be used to compute grasp quality metric ϵ, as explained above.

The ideal grasps executed in simulation for twelve objects from the validation set based on trained ResNet50 prediction along with the computed ϵ value is shown in Fig. 15.

6.3 Grasp Quality Comparison

Based on the value of ϵ, computed from simulation, the grasp quality between the trained ResNet50 model and GraspIt! can be compared. The grasp quality comparison over simulation was performed for the 49 objects from the validation set of EGAD dataset. For each object, ideal grasps were predicted using both the trained ResNet50 model and GraspIt!. While there were a total of 49 objects in the validation set, many simulated grasps were unstable due to slip, preventing the grasp quality calculation for these objects. As such, the comparison was only performed for objects (27 out of 49) that resulted in stable grasps under both the trained ResNet50 model and GraspIt!.

Figure 14 illustrates ideal grasps for different objects from the EGAD dataset, generated using the trained ResNet50 model and GraspIt! The side-by-side comparisons showcase stable grasps achieved on different objects from the EGAD validation set. Figure 16 provides a plot comparing the grasp quality between the trained ResNet50 model and GraspIt! over the 27 objects with stable grasps under both approaches. Among the 49 validation objects, the model achieved stable

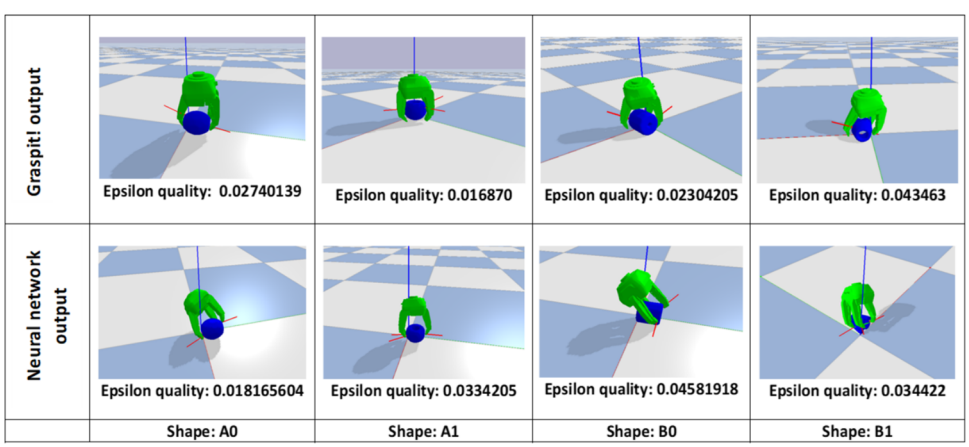

Fig. 14 Comparison between ideal grasps generated by GraspIt! and the trained ResNet50 Model for objects from the EGAD dataset. The ideal grasp and the epsilon quality metric for each object are shown

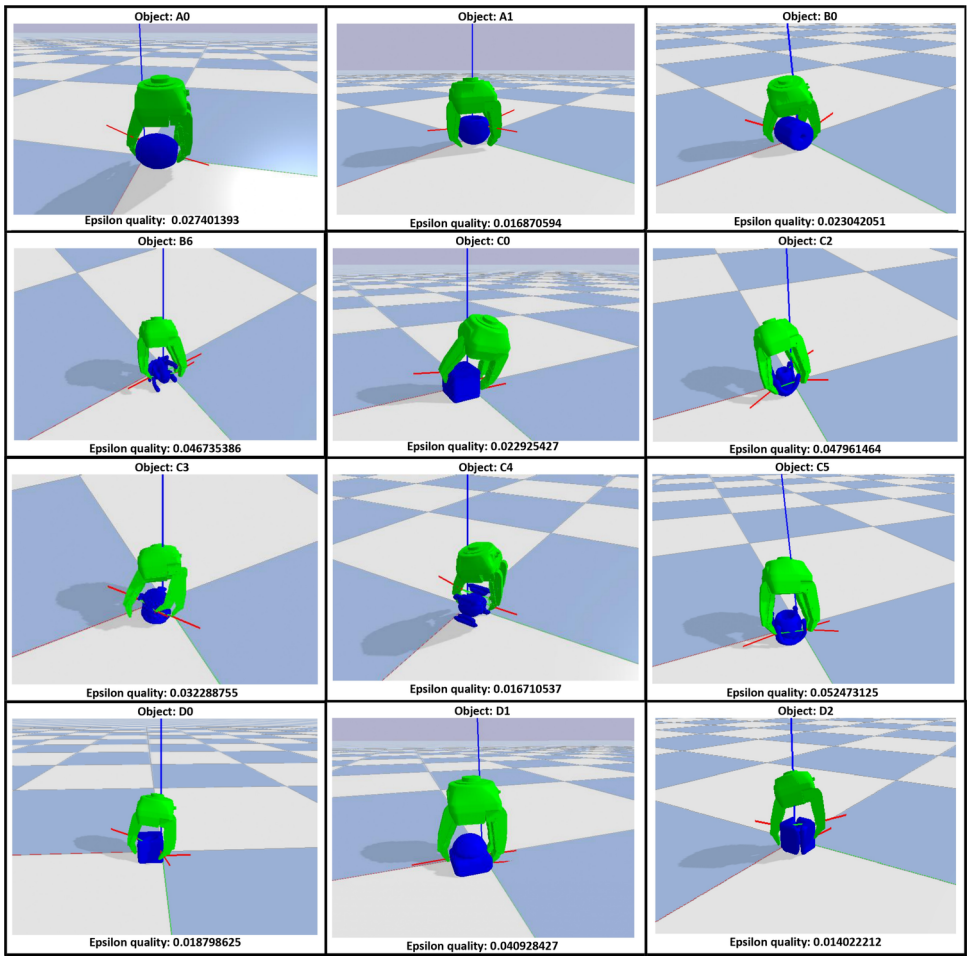

Fig. 15 PyBullet validation results: Each cell shows the grasp generated for a particular object as simulated in PyBullet along with its epsilon quality metric

grasps on 27 objects, with comparative performance in terms of grasp quality (Epsilon metric) by the ResNet50 model for several objects. It can be seen in Fig. 16 that out of the 27 stable grasps, 10 show higher grasp quality than the grasp generated from GraspIt!.

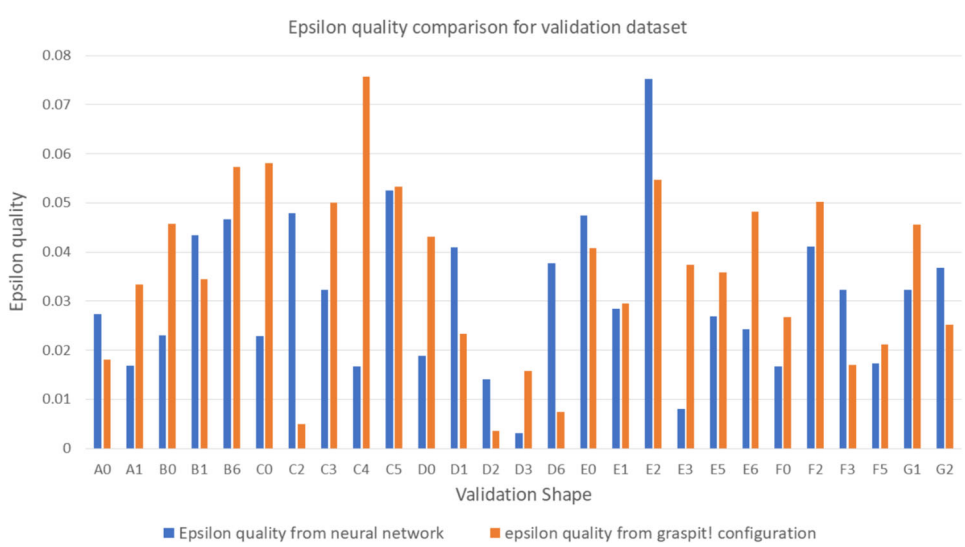

Fig. 16 Comparison of Grasp Quality (Epsilon Metric) between the trained ResNet50 Model and GraspIt! over 27 stable grasp scenarios

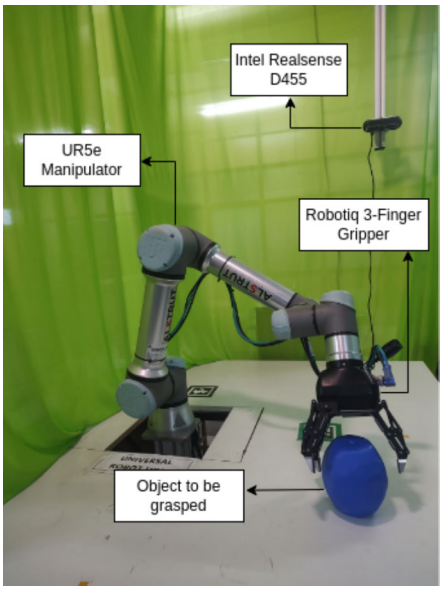

Fig. 17 Hardware setup for grasp validation: Consists of a Robotiq three-finger Gripper attached to a UR5e manipulator with an Intel Realsense D455 for RGB-D vision

In terms of inference time, the trained ResNet50 model offers substantial computational advantage. On average, the ResNet50 requires 15 ms per prediction. In contrast, the GraspIt! simulator takes around 20 seconds to compute the optimal grasp pose on the computer hardware (Intel Xeon CPU, 64 GB RAM, Nvidia RTX A4000 GPU with 16 GB VRAM). One major reason behind the faster computation is the GPU acceleration used by the trained ResNet50 model. GraspIt! does not support GPU acceleration. The substantial difference in computation time highlights the practical advantage of the proposed deep learning-based method for real-time robotic grasp planning applications.

The above results show that the grasps predicted by the trained ResNet50 model demonstrate the ability to handle a diverse range of objects while maintaining stability, comparable to GraspIt!'s performance. These results highlight the adaptability of the proposed representation, particularly for objects with varying shapes and complexities, demonstrating its applicability to real-world scenarios. The overall comparison indicates that while analytical methods like GraspIt! have long been a benchmark for robotic grasp planning, the trained ResNet50 model offers a viable alternative with the added advantage of faster computations and adaptability to new object geometries.

6.4 Hardware Validation

Hardware validation of the proposed grasp planning approach was performed using the setup shown in Fig. 17. The setup uses a 6-DOF UR5e robotic manipulator from Universal Robots for precise positioning of the gripper. A three-finger adaptive gripper from Robotiq was used to execute the grasps. Motion planning and trajectory execution for the manipulator were performed using MoveIt!. The MoveIt! framework is capable of computing collision-free trajectories for the manipulator, taking into account the gripper dimensions as well.

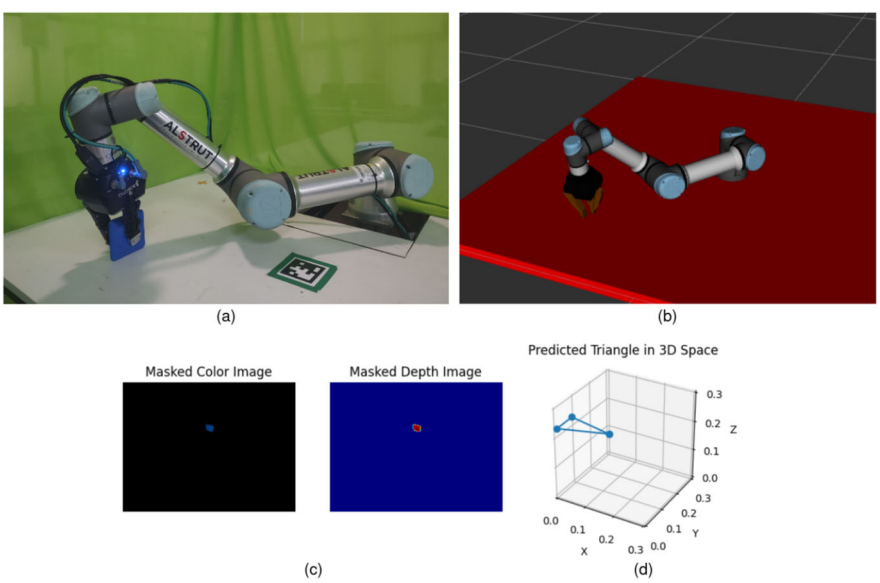

Fig. 18 Hardware validation: (a) Ideal grasps being executed using UR5e manipulator and Robotiq gripper (b) Motion plan generated by MoveIt! for the UR5e manipulator (c) Masked color and depth image as obtained from RGB-D camera (d) Ideal grasp as predicted by the trained model in proposed triangle representation

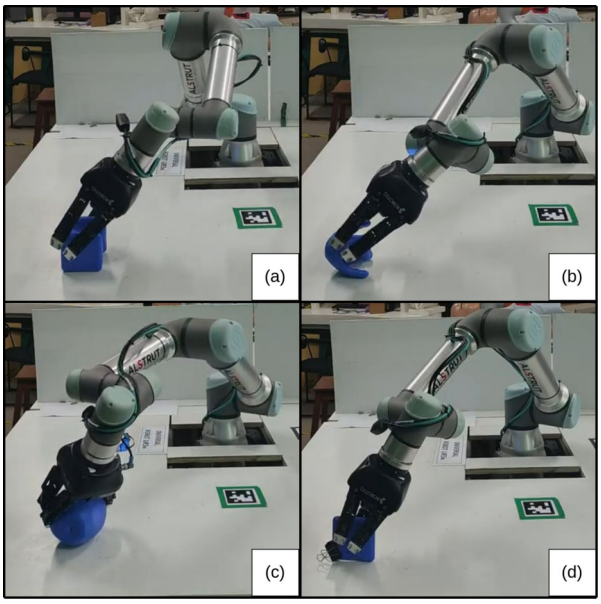

Fig. 19 Successful grasps achieved using the proposed network: Each cell shows a grasp on an object from the EGAD dataset

An Intel RealSense D455 RGB-D camera was used to capture the scene, providing color images and depth information. This information was used by the proposed approach for object detection, pose estimation, and grasp prediction. The overall grasp planning pipeline from RGB-D image to grasping the real-world object is shown in Fig. 1. A few successful grasps achieved by the network during the hardware validation are shown in Fig. 19.

Figure 18 shows one instance of the hardware validation. For an object placed within the field of view of the RGB-D camera, a masked color image and depth image are generated, which are then fed into the trained ResNet50 model. The model generates the ideal grasp points for the three fingers as a predicted triangle in 3D space. This represents the desired gripper position prior to initiating the grasp.

Based on the desired gripper pose, the MoveIt framework is used to plan a collision-free path from the current pose of the robotic manipulator. Once the gripper is moved to the desired pose by the manipulator through the planned path, the gripper is commanded to close around the object to complete a successful grasp as shown in Fig. 18. The videos of successful and failed trials under the hardware validation are provided in this repository [34]. The code for training and validation, as well as the trained model, is also made available in the above repository.

7 Conclusion

Robotic systems are being widely adopted to automate everyday tasks and routine chores at home and in professional settings, such as cleaning, sorting, food preparation, dispensing, and personal assistance. Traditional two-finger grippers are often not sufficient for the wide variety of grasps that may be required to effectively perform the above operations. This paper presents a novel triangular representation to better represent underactuated three-fingered grasps. A synthetic dataset was developed consisting of color and depth images of objects from the EGAD dataset, and the corresponding ideal grasps in the proposed novel representation. A modified ResNet50 architecture was trained on this dataset to perform end-to-end grasp planning.

The trained network was initially evaluated in simulation to compare the predicted grasp quality with GraspIt!. For the majority of objects from the EGAD validation dataset the grasp quality for the trained network was comparable to GraspIt! and in some cases better than GraspIt!. The trained grasp planning network was also successfully validated over real-world experiments, indicating good sim-to-real transfer capability under the proposed methodology.

While GraspIt! remains a powerful analytical tool, the trained network with the triangular representation offers several key advantages:

1. Computational Efficiency: The learning-based pipeline significantly reduces runtime from 20 seconds as taken by GraspIt! to 15 ms per grasp prediction. This makes the proposed approach suitable for real-time applications.
2. Robustness: The proposed method is less reliant on precise object models or exhaustive grasp synthesis, improving its robustness in dynamic and unstructured environments. GraspIt! and similar analytical approaches require a detailed 3D model of the object to perform grasp planning. Proposed approach works directly on RGB-D date obtained in real-time making it more robust for real-world applications, especially for previously unseen objects.
3. Adaptability: The triangular representation inherently captures the configuration flexibility of three-finger grippers, outperforming the traditional rectangular representations used for parallel-jaw grippers. This approach can be easily extended for n-fingered grippers using a polygonal representation.

Future work will explore the applicability of the proposed approach for grasping in cluttered environments and extending the approach towards grasping more complex shapes. Integration of existing slip detection and control mechanism [35] with the proposed approach would allow it to be used for handling delicate objects. While this work explored the use of a modified ResNet50 architecture for grasp planning, other architectures such as Pointnet[20] could be explored to allow for point-clouds and partial point-clouds as the input for grasp prediction.

GraspIt! has a strong theoretical background and offers good interpretability. It is a well established analytical method based on the epsilon metric and force closure, making it a consistent and explainable baseline for the evaluation of learning-based models for grasp planning, as demonstrated in this work. As part of future research, we plan to benchmark the proposed grasp planning approach against more recently developed state-of-the-art learning-based and hybrid grasp planning approaches, such as Dex-Net [11], GQ-CNN [12] frameworks that utilize deep neural networks for grasp prediction.

Supplementary information A provisional patent has been filed for the methodology described in this manuscript: System And Method For Prediction Of Grasping Position Of Robotic Gripping Mechanism On An Object; Application No: 202441057093

Author Contributions Every author contributed significantly to the research. Shreyash Gadgil is credited towards formulating the approach. Shreyash Gadgil and Prithvi Raj Baskar are jointly responsible for hardware validation. Bijo Sebastian, Alpana Dubey, and Abhinav Upadhyay gave the necessary guidance and support for the research. All authors contributed towards writing the manuscript.

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Data Availability The dataset, including PyBullet files, model architecture, and model weights presented in this manuscript are available in this repository [34].

Declarations

Conflicts of Interest The authors do not have any relevant financial or non-financial interests to disclose.

Ethics Approval The work does not involve human or animal participants, their data or biological material.

Consent to Publish The authors give their consent for the publishing of this article.

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